

SS1 & SS2 Chemistry — Umbrella Must-Knows

Following the NERDC/Lagos State Unified Scheme of Work, structured for the "Artificial Iman" pedagogical method

How to read this document

Every term's must-knows are sorted into two columns, because chemistry lies to students who don't know which kind of "fact" they're holding:

-  **CONSTANT** — a value or convention that exists because the universe (or a human committee) fixed it that way. No algebra gets you there. It must be *known*, cold, on recall, the way you know your own name.
-  **DERIVED** — a relationship that *follows* from reasoning, ratio, or proportion. It should never be memorized as a standalone fact — it should be re-derivable from first principles every time, even under exam pressure.

Where concepts are genuinely **the same entity described from a different axis** (your "3 names, same thing, different direction" pattern), they're grouped under a  **Same-Thing Cluster** — because teaching these as unrelated vocabulary is where most Nigerian chemistry students quietly start drowning.

This is the *umbrella* layer — the must-knows a student should carry in their head as non-negotiable before term-end. It sits above your reality-first teaching sequence; how you *arrive* at each must-know (ores vs brine vs crude oil, classification pressure, early derivation) is your lesson-delivery layer, not this document's job.

SS1

SS1 — First Term

Theme progression: Introduction to Chemistry → The Chemical World (Weeks 1-2: Intro & scientific method · Weeks 3-4: Matter, elements/compounds/mixtures, atomic structure · Weeks 5-9: Symbols/formulae/equations, periodic table (first 20), bonding, nomenclature · Week 10: States of matter/kinetic theory)

CONSTANT — must be known cold

- The three subatomic particles and their **relative charge and mass**: proton (+1, 1), neutron (0, 1), electron (−1, $\sim 1/1840$).
- **Atomic number = number of protons** — this is the identity anchor of an element; it does not change under any circumstance short of nuclear reaction.

- Electronic configuration of the **first 20 elements** (2,8,8 filling pattern) — recall, not derivation, at this stage.
- Standard **chemical symbols** for all common elements (Na, K, Fe, Cu, Pb, etc.) — arbitrary, historical, memorized.
- **Valencies of common elements and radicals** (e.g. SO_4^{2-} , NO_3^- , NH_4^+ , CO_3^{2-} , OH^-) — fixed conventions, not derivable from first principles at SS1 level.
- The three states of matter and their kinetic-theory particle arrangement (fixed vocabulary: fixed shape/volume, indefinite shape, etc.).

⚙️ **DERIVED — must be reasoned, never rote**

- Writing a **formula from valency** (cross-multiply rule) — this is algebra, not memory. A student who memorizes " MgCl_2 " instead of deriving it from $\text{Mg}^{2+}/\text{Cl}^-$ will fail the moment the compound is unfamiliar.
- **Number of neutrons** = mass number – atomic number — pure subtraction, re-derive every time.
- **Empirical vs molecular formula** relationship (molecular = empirical $\times$ n) — ratio reasoning.
- Balancing chemical equations — this is conservation-law algebra (atoms in = atoms out), never memorized as "the answer."
- Classifying element / compound / mixture from first principles (fixed composition? separable by physical means? single vs multiple substances?) — a reasoning test, not a lookup.

📦 **Same-Thing Clusters**

- **Atom $\rightarrow$ Isotope $\rightarrow$ Ion**: one entity (defined by proton number), three axes of change —
 - Isotope = same protons, **different neutrons** (mass axis)
 - Ion = same protons, **different electrons** (charge axis)
 - (Molecule will join this cluster in Term 2 as the **combination/bonding axis** — flag it now, resolve it once bonding is taught.)
 - **Element vs Compound vs Mixture**: not three unrelated categories but a spectrum of *how tightly matter is bound together* — element (one kind of atom) $\rightarrow$ compound (atoms chemically bonded, fixed ratio) $\rightarrow$ mixture (physically combined, no fixed ratio). The separation *technique* that recovers the original substances is diagnostic of which one you're looking at.
 - **Physical change vs Chemical change**: same event (matter transforming) viewed by whether the *identity* of the substance survives — this pairs directly with the element/compound/mixture cluster above.
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SS1 — Second Term

Theme progression: The Chemical World → Chemistry and Environment (*Weeks 1-3: Gas laws, mole concept intro · Weeks 4-6: Separation techniques · Weeks 7-10: Acids, bases, salts, water*)

CONSTANT — must be known cold

- **Avogadro's number:** 6.02×10^{23} per mole — a measured constant of the universe, not derivable.
- **Molar volume of a gas at STP:** 22.4 dm^3 (or $22,400 \text{ cm}^3$) — a fixed experimental value.
- STP conditions themselves: 0°C (273 K), 1 atm — convention, memorize.
- Color changes of standard indicators: litmus (red↔blue), phenolphthalein (colorless↔pink), methyl orange (red↔yellow) — arbitrary chemical facts, not derivable.
- The general word/symbol pattern: acid + base → salt + water — a named relationship (definition), not an equation to "solve."

DERIVED — must be reasoned, never rote

- **All gas law relationships** (Boyle's, Charles's, Gay-Lussac's, general gas equation) — these are proportionality statements. A student should be able to re-derive "if temperature increases, pressure increases" from kinetic theory reasoning, not memorize it as an isolated law.
- **Moles = mass ÷ molar mass** and **moles = volume ÷ molar volume (gases)** — these are unit-conversion algebra, always re-derivable from dimensional reasoning, never "just known."
- Choosing the correct **separation technique** for a given mixture — this is a diagnostic/reasoning skill (what property differs? boiling point? solubility? particle size?), not a memorized pairing.
- Predicting strong vs weak acid/base behavior from degree of ionization — reasoning from a definition, not a lookup table.

Same-Thing Clusters

- **Acid / Base / Salt:** not three separate topics but one neutralization system viewed from three positions — acid (H^+ donor) and base (OH^- or proton acceptor) are the two reactive poles, salt is the *structural residue* left once they cancel. Every salt "remembers" which acid and which base made it (hence acid salts, basic salts, normal salts).
- **Evaporation / Boiling / Distillation / Crystallization:** same underlying phenomenon — *phase change driven by energy input* — differing only in what's being separated and how much energy is controlled. Teaching these as four unrelated "methods" instead of one energy-driven family is a common source of confusion.

- **Hardness of water (temporary vs permanent):** same cause-category (dissolved $\text{Ca}^{2+}/\text{Mg}^{2+}$ salts) split by *which anion* is present (bicarbonate vs sulphate/chloride) — this determines whether boiling alone fixes it.
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SS1 — Third Term

Theme progression: Chemistry of Life → Chemistry and Environment → Chemistry and Industry (Weeks 1-7: *Carbon and its compounds — allotropes, coal, coke, CO_2/CO , trioxocarbonate, hydrocarbons* · Weeks 8-10: *Chemical industries*)

CONSTANT — must be known cold

- The **three allotropes of carbon** (diamond, graphite, amorphous/charcoal forms) and their fixed structural facts (diamond = 4 bonds/tetrahedral, graphite = 3 bonds/layered) — structural facts, not derivable.
- Products of **complete combustion** ($\text{CO}_2 + \text{H}_2\text{O}$) vs **incomplete combustion** ($\text{CO} + \text{H}_2\text{O}$, or soot) — a fixed chemical outcome dependent on oxygen availability, memorized as a pattern.
- Formula and toxicity mechanism of **carbon monoxide** (binds hemoglobin) — a biological/chemical fact, not derived.
- Classification: **heavy chemicals vs fine chemicals** in industry, and named categories (fertilizers, plastics, pharmaceuticals, metallurgy) — industrial/economic categories, memorized.

DERIVED — must be reasoned, never rote

- Why diamond is hard and graphite is soft — this must be reasoned from bonding structure (tetrahedral rigid lattice vs layered sheets with weak interlayer forces), never memorized as an isolated fact disconnected from bonding.
- Writing and balancing the equation for a carbonate + acid reaction (test for CO_2) — algebra from conservation.
- Reasoning why gas flaring wastes hydrocarbon fractions economically — an applied-logic exercise linking petroleum composition to industrial value.

Same-Thing Clusters

- **Carbon allotropes:** same element (carbon, proton number 6), different **bonding architecture axis** — this is the first place students should notice that "same atom, different arrangement" is its own axis of variation, distinct from the isotope/ion axes from Term 1. By the end of SS1, the full cluster is: *same atom → different mass (isotope) → different charge (ion) → different bonding arrangement (allotrope/structure)*.
- **Coal / Coke / Charcoal:** same general "carbonized organic material" category, differing by *source* and *degree of processing* — coal (fossil, geological), coke

(industrially processed from coal), charcoal (from wood) — same functional role (carbon fuel/reducing agent), different origin stories.

SS2

SS2 — First Term

Theme progression: The Chemical World (periodicity, redox, electrochemistry) (Week 1: Periodic table history/blocks · Weeks 2–3: Families of elements, periodic trends · Weeks 4–5: Redox reactions, oxidation numbers · Weeks 6, 8–9: Electrolysis · Weeks 10–11: Electrochemical cells)

CONSTANT — must be known cold

- The **8 groups / 7 periods** structure and the **4 blocks** (s, p, d, f) — this is the fixed architecture of the periodic table, memorized as a map before it's reasoned about.
- Names and general reactivity character of each **family**: alkali metals, alkaline earth metals, halogens, noble gases, transition metals — named categories with characteristic (memorized) behavior.
- **Rules for assigning oxidation numbers** (e.g. O is usually -2 , H is usually $+1$, uncombined element = 0) — these are conventions, agreed rules, not derivable from physics at this level.
- **Faraday's constant** and Faraday's 1st and 2nd Laws as stated relationships — the laws' existence and statement are given; only the calculation *application* is derived (see right column).

DERIVED — must be reasoned, never rote

- **Periodic trends** (atomic radius, ionization energy, electronegativity across a period/down a group) — these must be reasoned from nuclear charge + shielding + shell number, every single time, never memorized as "trend arrows" without the electronic reasoning underneath.
- **Calculating oxidation numbers** in a compound or ion — this is algebra: the sum of oxidation numbers equals the overall charge. Apply the rule, solve the equation.
- Identifying oxidizing vs reducing agent in a given reaction — reasoning from electron transfer direction (OIL RIG), not lookup.
- **Faraday's Law calculations** (moles of electrons $\leftrightarrow$ mass deposited $\leftrightarrow$ quantity of electricity, $Q = It$) — once the law is known as a *given*, the calculations built on it are pure algebra/dimensional reasoning.
- Predicting **preferential discharge** of ions at electrodes — reasoning from reactivity/concentration factors, not memorized per-compound.

Same-Thing Clusters

- **Oxidation / Reduction:** not two separate reactions but **one electron-transfer event seen from two ends** — whatever loses electrons is oxidized, whatever gains them is reduced, and they always happen together (redox). This is the single most important "same thing, different direction" relationship in SS2 — teach it as one coin, two faces, not two topics.
 - **Electrolytic cell vs Electrochemical (galvanic) cell:** same underlying apparatus-logic (electrodes + electrolyte + electron flow) but **opposite energy direction** — electrolytic cells consume electrical energy to force a non-spontaneous reaction; electrochemical cells release electrical energy from a spontaneous reaction. Same components, inverted causality — this is a second "different axis" pattern worth naming explicitly.
 - **Molecule** rejoins the atom/isotope/ion cluster here, now that bonding is fully built: atom (neutral baseline) → isotope (mass axis) → ion (charge axis) → molecule (combination axis, specifically *covalent* combination) → and now this term adds the **electron-transfer axis** implicit in redox, which is really just ionic bonding "caught in the act."
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SS2 — Second Term

Theme progression: Chemical Reactions, Energy/Equilibrium, Non-metals (H, O, Halogens, N, S) (*Weeks 1-2: Rate of reaction, collision theory · Weeks 3-4: Energy changes, thermodynamics, entropy/Gibbs · Week 5: Equilibrium · Weeks 6, 8-11: Hydrogen, Oxygen/air, Halogens, Nitrogen, Sulphur*)

CONSTANT — must be known cold

- **Position on the periodic table** for H, O, halogens, N, S — fixed facts (group/period), memorized as location data.
- **Percentage composition of air** (~78% N₂, ~21% O₂, trace CO₂ and noble gases) — a measured constant of the atmosphere.
- Named **allotrope of oxygen** (ozone, O₃) — a fixed fact.
- The named **oxides of nitrogen** and their formulas (NO, NO₂, N₂O) — fixed nomenclature/formula facts.
- Conditions of the **Haber process** (N₂ + 3H₂ ⇌ 2NH₃, iron catalyst, ~450°C, ~200 atm) — an industrially fixed set of conditions, memorized.
- Test reactions/color changes for halides, ammonia, nitrate ions — fixed observational facts.

DERIVED — must be reasoned, never rote

- **Rate of reaction and the four factors affecting it** (concentration, temperature, surface area, catalyst) — each must be reasoned through collision theory (frequency and energy of collisions), never listed as disconnected facts.

- **Activation energy and energy profile diagrams** — reasoned from bond-breaking/bond-forming energy balance.
- **Le Chatelier's Principle applications** — this is a reasoning tool (predict shift direction from the stress applied), not a memorized table of outcomes. Every equilibrium shift question should be re-derived from the principle, live.
- ΔG , ΔS , ΔH relationships ($\Delta G = \Delta H - T\Delta S$) and spontaneity conditions — algebraic reasoning from the relationship, not memorized conclusions.
- Explaining **gradation of properties down Group VII (halogens)** — reasoned from increasing atomic size/decreasing electronegativity, not memorized as isolated per-element facts.

Same-Thing Clusters

- **Exothermic / Endothermic**: one energy-transfer event, two directions — heat *released* vs heat *absorbed*, both defined relative to the surroundings. This should be taught as a single axis (energy flow direction) with two labeled ends, exactly like oxidation/reduction was a single axis with two labeled ends.
- **Reversible reaction / Equilibrium / Le Chatelier's shift**: same system, three levels of description — reversible (the reaction *can* go both ways) → equilibrium (the *state* where forward rate = backward rate) → Le Chatelier (the *rule* for predicting how that state moves when disturbed). Not three topics — one system, described at rest, then in motion.
- **Hydrogen / Oxygen / Halogens / Nitrogen / Sulphur**, taught back-to-back this term, all share one underlying question-template worth making explicit to students: *position* → *electronic configuration* → *property gradation* → *lab prep* → *industrial prep* → *test*. Once a student sees this is the *same investigative pattern* applied to five different elements, the "memorize 5 separate topics" burden collapses into "apply 1 pattern 5 times."

SS2 — Third Term

Theme progression: Water, Solubility, Mass-Volume Relationships, Volumetric Analysis, Hydrocarbons, Alkanols (Weeks 1-2: *Water* · Week 3: *Solubility* · Week 4: *Mole/mass-volume* · Weeks 5-6: *Volumetric analysis/titration* · Weeks 8-11: *Hydrocarbons — alkanes, alkenes, alkynes, alkanols*)

CONSTANT — must be known cold

- **Molar volume at STP = 22.4 dm^3** (recurrence from SS1 — now applied at calculation speed, still a fixed value).
- **General formulae** of the hydrocarbon series: alkanes $\text{C}_n\text{H}_{2n+2}$, alkenes C_nH_{2n} , alkynes $\text{C}_n\text{H}_{2n-2}$ — these are *definitions* of the series (fixed pattern conventions), memorized as anchors even though individual formulae are then derived from them.

- IUPAC naming *conventions* (prefixes meth-, eth-, prop-, but- for chain length; suffixes -ane/-ene/-yne) — arbitrary naming rules, memorized.
- Bromine water test result for unsaturation (decolorization = alkene/alkyne present) — a fixed observational fact.

⚙️ **DERIVED — must be reasoned, never rote**

- **Titration calculations** ($C_1V_1/n_1 = C_2V_2/n_2$ mole-ratio method) — pure stoichiometric algebra, re-derived from the balanced equation every time, never memorized as a formula to plug into blindly.
- **% purity/impurity, concentration in g/dm³ and mol/dm³, water of crystallization calculations** — all downstream algebra from the mole concept, not separate memorized formulas.
- **Solubility curve interpretation** and calculations — reading a proportional relationship, reasoning not recall.
- Naming an unfamiliar alkane/alkene/alkyne from structure, or drawing structure from name — this is applying the general-formula *pattern*, never memorizing individual compound names as a list.
- Predicting chemical behavior of alkanes vs alkenes (substitution vs addition reactions) — reasoned from saturation/unsaturation, not memorized per-compound.

📁 **Same-Thing Clusters**

- **Solute / Solvent / Solution / Saturation states:** one system (a substance dissolving in another) viewed across a **concentration axis** — unsaturated → saturated → supersaturated is a single continuum, not three categories.
- **Alkanes / Alkenes / Alkynes:** same "hydrocarbon" identity, differing along **one axis** — **degree of saturation** (number of C-C bonds shared between neighboring carbons: single, double, triple). This is the same "one entity, different axis" pattern as isotope/ion, just applied to bonding multiplicity instead of mass or charge — worth explicitly telling students so they recognize the pattern-of-patterns.
- **Titration / Volumetric analysis / Standard solution:** not separate procedural vocabulary but one measurement system — a *standard solution* is the trusted reference, *titration* is the technique that compares an unknown against it, *volumetric analysis* is the umbrella name for the whole measurement discipline.
- **Hardness of water** (reintroduced from SS1) now gets its quantitative resolution here via solubility and volumetric analysis — flag this explicitly as "the SS1 qualitative story now gets its SS2 calculation."

Cross-year "axis" reference (the running thread)

By the end of SS2, a student should be able to recite, unprompted, that the following are all instances of **one pattern** — *same fundamental entity, described from a different axis of*

variation — and name which axis each one is:

Entity pair/cluster	Axis of variation
Atom → Isotope	Mass (neutron count)
Atom → Ion	Charge (electron count)
Atom → Molecule	Combination (bonding)
Carbon → Allotropes	Structural arrangement
Oxidation ↔ Reduction	Direction of electron transfer
Exothermic ↔ Endothermic	Direction of energy transfer
Electrolytic cell ↔ Electrochemical cell	Direction of energy conversion
Alkane → Alkene → Alkyne	Degree of saturation (bond multiplicity)
Unsaturated → Saturated → Supersaturated solution	Concentration relative to solubility limit

This table is the spine. If a student can independently generate this table from memory by SS2 third term, the pedagogy has worked — because it means they've stopped seeing chemistry as a pile of vocabulary and started seeing it as a small number of axes wearing different costumes.

Sources: Lagos State Ministry of Education Unified Schemes of Work for Senior Secondary Schools 1 & 2 (Chemistry), cross-checked against the NERDC-aligned scheme summaries independently published for SS1/SS2 chemistry term breakdowns.